

Course Notes — SESSION 7

INTERACTION

1. Introduction

Let us summarize our story so far: AI systems and organizations are intelligent if they can learn to assign credit to actions when solving complex problems. They assign credit by manipulating symbols. Symbols are units such as bit strings and lists, or composites of these such as search trees, schemas, or constraint networks. Manipulations correspond to processing information about a problem-solving environment, or to computing the steps of a program. Different AI ideas or paradigms (e.g., reinforcement learning versus learning-in-a-society; search versus constraint propagation) are essentially differences in strategies for manipulating symbols.

Diverse researchers in the AI field find this story wrong or misleading. The story suggests intelligent problem-solving is all about **internal cognition**, or thinking about problems in the world *separate from the external world itself*. Such a view of intelligence goes back at least to the 17th century and the French philosopher **René Descartes**. His phrase, “**I think, therefore I am**” implies that the very existence of intelligence is constituted by internal symbol manipulations. In the view of these diverse researchers, all symbol manipulation approaches to studying AI systems and organizations are repeating the same mistakes as narrow **Cartesianism** (also called **cognitivism**) approaches of not dealing adequately with the external world:

Most AI ideas are stuck in the same narrow Cartesian approaches as 400 years ago.

These diverse researchers propose that AI systems and organizations be designed based on identifying agents’ *interactions* with the *external physical and cultural world*. The term **interaction** is used to emphasize roles beyond internal cognition in intelligence of **sensorimotor action** (perceptual senses and physical movements) and **experiencing phenomena** in the world. Building a “model” of the world symbolically in programs or databases neglects the power of these forces of intelligence, and needlessly complicates and bureaucratizes things. Instead, AI systems and organizations can learn to assign credit more efficiently and effectively by just directly interacting with it. This interactionist point of view is nicely summed up by a famous phrase of roboticist and interactionist AI researcher Rodney Brooks:

“The world is its own best model”.

Consider our previous session’s example of Sudoku. Google’s Peter Norvig claims to have designed a Python program that solves any Sudoku puzzle. Is this claim plausible? From an interactionist approach to AI, it is misleading. Norvig’s Python program depends on the computer being turned on, on whoever runs the program to be aware of what Sudoku is, and so forth — an entire system able to interact with the physical and cultural world. In this session, we

will explore how we might understand and design AI systems and organizations based on foregrounding such interactions, and not just symbol manipulations.

2. Defamiliarizing our language for talking about AI

The basic AI idea described above seems straightforward enough: focusing on interaction with the external world, not just internal symbol manipulation. Yet researchers of interactionism have found that understanding this approach demands of us a major change in perspective. Symbols are easily noticed, such as the text in these course notes. Paradoxically, our interactions with the world are so ubiquitous that they easily go unnoticed. Cartesianism has dominated scientific thought for centuries. Most of our **language for talking about intelligence** reflects its focus on internal cognition — *Cartesianism has become the water that we are swimming in*.

Consider how easily we thought of Sudoku playing as based on constraint propagation, while putting aside our physical actions or experience of the game. Likewise, in organizations, going back to at least Babbage, the dominant view has been industrial, in which economic activities can be divided into symbolic tasks and managed through the symbol manipulation processes of a bureaucratic administration. The interactionist argument is that, to see how all this may not, in fact, be so intelligent, what is needed is to **defamiliarize** our Cartesian *language* that we have inadvertently been using to talk about intelligence. We explore how to do such defamiliarization in terms of our Cartesian *metaphors of action*, our *stories about learning*, and *concepts of complexity* in organizations and other systems.

2.1 Agre's critical technical practice: defamiliarizing Cartesian metaphors in AI

In pioneering work in the 1980s at MIT, the AI scholar Phil Agre introduced a method for surfacing what he saw as the narrow yet vague language of AI ideas based on symbol manipulation that he called “**critical technical practice**” (CTP). The CTP method involves (1) surfacing assumptions in the language of AI ideas, (2) evaluating assumptions with respect to observations and experience of concrete phenomena in the world, and (3) using the evaluations to guide the construction of alternative AI ideas and systems. We will focus in this section on the first aspect of CTP of surfacing assumptions.

Linguists have pointed out that much of our everyday language relies on metaphors with the physical world. For example, we say that we “grasp” ideas, “wrestle” with problems, or “move forward” with plans”. Agre proposed that how such metaphors are used in the AI field is often harmful in that it casually conflates internalist (Cartesian) symbol manipulation with actual real-world concrete phenomena. Consider the terms that we have freely used so far in this course to describe what happens in AI systems, such as *learning*, *planning*, *searching*, *reasoning*, or *having goals* or *knowledge*. Is how we described these terms really like how they actually happen in real-world activity? “Searching” for your keys might involve looking around your house or

feeling through your pockets, which is not really like searching through a “tree” of symbolically-represented states. Planning in organizations does not really resemble storing sequences of actions in such a tree.

Instead, Agre suggests, such activities are better thought of as **improvised** through our moment-to-moment interactions for “coping” with the world rather than representing it. For example, an organization’s “plans” for launching a new product are continually negotiated based on interacting with the physical product itself, the physical production environment, and with customers to see how they experience and use it. Ignoring such improvisations is tempting, since it enables us to understand search, planning or other terms solely in terms of well-defined symbol manipulations. They serve as what have been called “**wishful mnemonics**” — memorable (mnemonic) terms for symbol manipulation processes that allow us to “wish” that such processes actually capture real-world phenomena. The danger is that such metaphors become a self-fulfilling prophecy: we may start thinking about search, planning (or learning or knowledge) as what happens symbolically inside of a computer. According to Agre, this will unnecessarily constrain our design of both AI systems and organizations.

2.2 Brooks’ “creature hypothesis”: defamiliarizing Cartesian stories about learning

The AI field has often set the benchmarks for intelligent learning in terms of individual Cartesian-style tasks for manipulating symbols. An early benchmark for AI was to learn to play chess like a grandmaster; more recent benchmarks include learning the game of Go or passing multiple choice tests. Whereas progress on such “complex” benchmark tasks has in some cases been impressive, seemingly simpler tasks that involve interaction with the physical and cultural world (e.g., getting a mouse to navigate a real-world environment, and not just a symbolic representation of a maze) have proven much harder than had been expected. This is called **Moravec’s paradox**, after Carnegie-Mellon roboticist Hans Moravec:

Moravec’s paradox: seemingly complex problems in AI are often simpler than expected; seemingly simple problems in AI are often (much more) complex than expected!

Roboticist Rodney Brooks proposed a “**creature hypothesis**” of intelligent learning that explains the paradox from an interactionist point of view. The creature hypothesis is based on the observation that humans developed language for symbolically representing the world only recently in evolutionary time (*thousands* of years ago) — and only after *millions* of years of adaptation directly with the physical world that originated in unicellular organisms and evolved many organisms (“creatures”) prior to humans. This observation suggests that it takes a lot of evolutionary time to develop the ability to do seemingly simple sensorimotor actions (e.g., to go from a unicellular organism to even a primitive animal). In contrast, once primitive humans developed certain sensorimotor and experiential abilities (e.g., standing on two feet, using thumbs), then language developed rapidly. Hence, counterintuitively, the simple sensorimotor

interactions with the physical and cultural worlds are the “hard” part of intelligence, while language and representation are the easier part.

For Brooks, one consequence of a Cartesian view of learning is a recurring theme in the AI field of **extrapolation** — of believing that a large gain in performance on one type of task (e.g., playing Go) means that the field is close to solving most other types of task and, hence, that artificial general intelligence (AGI) is imminent. The danger is that such extrapolation leads to large shifts in attention and resources to whatever the current paradigm is (e.g., LLMs today), while all the actually complex parts of intelligence (as laid out in the creature hypothesis) run the risk of never getting addressed.

2.3 Wiener’s “variety”: defamiliarizing Cartesian concepts of complexity

Cartesianism implies a concept of complexity in terms of the amount of internal symbol manipulations. We saw measures of such complexity as, for example, Shannon entropy or the number of steps to run a program in a Turing machine. Consider how these concepts derive from, or relate to, the idea of an industrial division of labor, as characterized by our pin factory example, into isolated tasks that can be described symbolically.

From an interactionist point of view, such derivations reflect a view of economic activity, and systems in general, that is far too internalist and industrial — e.g., the pin factory is fully characterized by the description of symbols about it. Such internalism reflects Cartesianism’s close affinity with Newtonian mechanics — a way of describing the world in terms of gear-like metaphors (e.g., as individual tasks that map neatly to physical components). In this sense, Simon’s watchmaker parable neatly fits the picture: it continues to draw on mechanical clocks as a canonical example of a complex system that has been around since Descartes’ time. AI researchers from an interactionist point of view argue instead that concrete, real-world phenomena are far too complex to capture complexity in terms of such internal symbol manipulations. Even in an industrial factory, the amount of complexity is vastly higher than one would be led to believe by Babbage’s concept of a division of labor or the Turing machine.

Going back even before the formal emergence of the AI field, the MIT mathematician and polymath Norbert Wiener proposed a concept of complexity in terms of *variety*, where variety referred to a level of complexity more reflective of natural systems (biological organisms or human societies and organizations) than internal symbol manipulations. Wiener’s view has influenced many interactionist researchers in AI to argue that what intelligence requires is a “**non-modern**” (e.g., non-Cartesian, non-Newtonian, or non-industrial) concept of complexity that foregrounds novel ways of organizing direct interactions with the physical and cultural world rather than of structuring symbol manipulations.

3. Criteria for intelligent interaction

3.1 Goals: homeostasis or “comportment”

The earliest ideas about interaction in AI originated in the field’s immediate predecessor called cybernetics, which arose in the 1940s led by Wiener. “Cyber” means “to govern”, and cybernetics means “science of control”. In cybernetics, control comes from feedback (i.e., information) from interacting with the environment. Perhaps the most canonical example of interaction in cybernetics is an analog **thermostat** in a room. The thermostat is set to maintain the temperature of the room within a certain range; if it gets too cold, the thermostat is triggered and the heater starts. This canonical example illustrates the first major concept in AI ideas of interactionism, which is an emphasis on goals of **maintaining a state**, rather than of optimizing or satisfying a specific goal within a state. This idea of maintaining a goal was termed by the cybernetician **W. Ross Ashby** as **homeostasis** (“same state”).

In the early 1950s, another cybernetician **W. Grey Walter** designed a “**tortoise**” robot that used a small set of simple analog circuits to navigate around a room. Consider the difference from Shannon’s Theseus labyrinth — rather than learn a specific path through a labyrinth, Walter’s tortoise was *homeostatic* in that it learned how to wander around the far less structured environment of a house and find its way back from anywhere in the house.



W. Grey Walter’s tortoise robot (1951)



Brooks’ Roomba vacuum cleaner that navigates any real-world home environment

In the 1980s, the MIT roboticist Rodney Brooks built a series of robots inspired by Walter’s ideas, which led to developing a commercial robot vacuum cleaner called the Roomba. It remains easily the best-selling robot of all time, with more than 40 million sold. Brooks’ approach differed from the prior paradigm in robotics that was based on search and representation (and from today’s dominant paradigm, which is based on deep learning and reinforcement learning using vast volumes of data). Brooks’ approach with the Roomba succeeded based on its simplicity — the Roomba had no CPU, no complex algorithms, and no vast stores of data. Rather, it used only a small set of simple circuits to learn to react to any type of room by wandering around. In addition to the ability to learn to navigate a new physical environment, the other advantage of Brooks’ approach was that “simple” meant far less costly. Making super cheap robots enabled new strategies for using robots. For example, Brooks and his

colleagues used similar ideas of simplicity to design large numbers of cheap robots for exploring Mars, different from the prior strategy of designing one complex and expensive robot. Whereas if a single expensive robot broke down or got stuck, the mission could be a failure, Brooks' "army" of robots tolerated the failure of potentially many such robots.

Overall, in Cartesian approaches to AI, the goals were to optimize or satisfy solutions, or satisfy a set of constraints with respect to a problem with a well-defined goal state. In the AI idea of interaction, the goal is instead about maintaining some sort of alignment with the physical situations in the world as they unfold. Another term for this homeostasis is "**comportment**", which means maintaining an authentic alignment between our goals and the physical world based on our moment-to-moment coping. When an organization is skillfully "comported" to the physical world, its members have become so familiar with it that most of the environment becomes just background. Skillful could mean a specialized skill, such as a soccer team that seems to play together effortlessly, or our everyday common sense, such as how a team of medical professionals are able to navigate their way through cases mostly without consciously thinking about it.

3.2 Action as situated, embodied, and embedded

The above examples suggest important aspects of intelligently solving problems are learned using sensorimotor actions to interact with and experience the physical world, and where goals are *homeostatic* — about maintaining states, rather than finding a single optimal or satisfactory solution to an isolated problem. Before we go into specific AI ideas about such interactions, let us try to make sense of what types of actions are required for homeostasis and comportment. What is it about interactions that enable intelligent credit assignment? Interactionist scholars of AI and organizations understand intelligent action and problem-solving as **situated**, **embodied**, and **embedded**. Understanding these characteristics will help frame for you why such scholars consider representations to be less fundamental to intelligence.

- **Situatedness.** Intelligent action is **situated** in that it depends on sensing and responding to ongoing changes in particular situations in the world. Solving complex problems requires not just being able to solve, for example, every kind of maze or possible chess puzzle. True generalization requires the ability to select the *relevant* characteristics of *real-world physical environments* in which actions are performed. In medical diagnosis, for example, beyond just adapting treatments for particular patients, situatedness also refers to the need to deal with all aspects of physical environments in which diagnosis actions are performed (e.g., medical facilities, rooms, equipment, etc.).
- **Embodiment.** Intelligent action is **embodied**, meaning that it depends on the ability of a body to learn physical actions, not just the mind's ability to learn symbols. For example, "thinking" about and planning a route from KAIST to the airport in a Cartesian manner presumes our physical ability to move our bodies and sense where we are. Embodiment is important in that it emphasizes limits to intelligence that are not like, e.g., Shannon limits on information, but rooted in limits in the physical motions that our bodies can perform.

For example, even if we expand our “channel capacity” infinitely for processing symbolic information, our ability to go from KAIST to the airport is still limited by how we can only take one step at a time with our feet.

- **Embeddedness.** Finally, intelligent action is **culturally embedded**, meaning that it depends on the local culture in which it takes place, which changes the **goals** of intelligent action. For example, taking the train to the airport in Korea depends on obeying laws and norms that may vary from other countries, such as no eating and very little talking. As another example, a crosswalk in Korea is still something that must be crossed by a pedestrian with caution (since cars will tend to go through the crosswalk, even given a waiting pedestrian); in contrast, in Vietnam, even if there is no crosswalk (and a lot of traffic), it is often acceptable just to walk across the street without waiting.

3.3 Assigning credit efficiently: the world as its own best model

Brooks’ phrase “**the world is its own best model**” (where by “model”, he was referring to the use of Cartesian-style symbolic representations) implies that an AI system or organization can be designed based on relatively simple **analog** mechanisms, such as the simple circuitry of the Roomba that enable homeostasis. Hence, an intelligent AI system or organization could be designed *without* any digital computers or even any other typical mechanism for symbol manipulation.

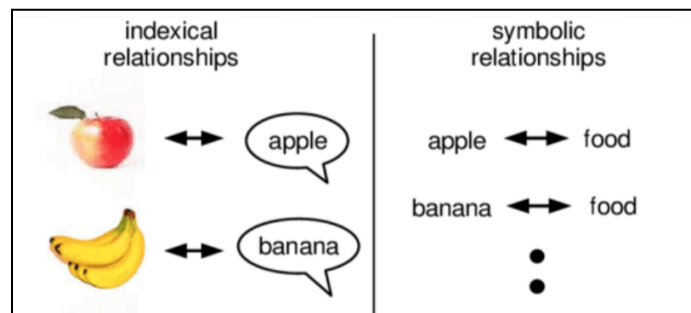
Nonetheless, there is nothing in interactionist views of AI that is *against* representations of symbols per se. Interactionist researchers in AI do not deny that representations can be important for intelligence. The claim is that symbolic representations are always secondary to more fundamental mechanisms of moment-to-moment improvisation. Representations become significant in particular moments when our homeostatic comportment fails, referred to as “**breakdowns**”. For example, you might walk through a building without giving it much thought, but then, when encountering a door that you are not familiar with, you stop and think about how to open it. Opening the door is clearly important for navigating the building; yet it is secondary to the moment-to-moment navigation actions. In this sense, representations are thought of as potentially valuable **resources for action**:

Symbolic representations are not necessary for interacting with the world; they are resources for action.

To consider the implications, first consider how this “resources for action” view compares to our earlier notion of Simon information and its emphasis on **designation** — that intelligence requires symbols that “designate” objects in the actual world by giving them names. For AI scholars of interaction, *designation is not enough* — it only accounts for the real-world meaning of symbols (i.e. lists) for a *given problem*, and not to the holistic experience of a

specific situation as it unfolds. We next consider two interactionist alternatives to designation — called “indexing” and “registration” — that seek to reframe symbols and symbol manipulation as grounded in our experience of real-world situations.

Indexing. In this view, developed also by Phil Agre at MIT, objects in the real-world are “indexed” in an agent’s memory, which means that they are stored *with respect to a specific situation*. This is like a word in an index of a book: it does not *just* have a name; the word also has a page number that “points” you to the specific “situation” in the specific book to which its meaning relates. To make our comparison of designation to indexing still sharper, consider how the name “apple” can be designated, for example, also by “pointing” to another related symbol such as “food”. Yet, this is just a list structure (as Newell & Simon had in mind in their physical symbol system hypothesis) — one name points to another name. Indexing, in contrast, refers to using “apple” to refer to a *particular physical object* (a particular apple) in the world, just as “banana” could be used to point to a particular banana.



As a broader example, think of the chair of the room at which you commonly study. You not only know this object by its name (“that is a chair”), but by how it is related to your overall study room. Hence, it is more accurate to your experience of it to think of it when you see it in the room as “*that-chair-right-there*”. In terms of AI and organizations, the key idea is that indexing the objects in the physical world allows them to become familiar to us, such that our sensorimotor actions can be far more efficient and real-time than if we had to “plan” all of them (e.g., by processing information in lists). Consider, for example, how when you study in your study room (or walk down the street, take the bus, brush your teeth etc.), the vast majority of objects in the physical world are so familiar to you that you do not have to think much. In this way, our embodied and embedded interactions with situations in the world can be thought of as ongoing **improvisations** made possible by having indexed the objects in the world around us according to how we experience them.

Registration. In a complementary theory proposed also at MIT by AI researcher **Brian Cantwell-Smith**, AI systems need to engage in the “registration” of symbols, where registration refers to a process of understanding or making sense of coherent physical and cultural worlds in which we are involved. In registration, representations are various **schemes**

— collections of symbols that cohere and make sense to us. Examples of schemes could be anything from blueprints, to a database table, to a story. “Registration” refers to accounting for (i.e., “registering”) (1) whether these schemes accurately reflect our experience of the world; (2) whether our experience of the world is in fact accurate to the “reality” of the world. For example, to design a building, an architect may have some vision for its form and function. During the design process, the architect works with various schemes (blueprints, building codes, images of similar buildings, etc.), and visits the actual building site. During this process, the architect may realize that either that one or more of the schemes are wrong, or that the actual site is wrong for the initial vision for the form and function of the building. It is this ongoing interplay between representations (schemes), reality (the physical world) and our experience of these — what Cantwell-Smith calls the interplay between “reckoning and judgment” — that refers to processes of registration.

4. Strategies for interaction in AI and organizations

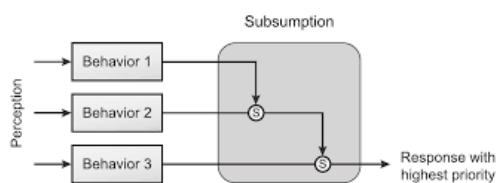
The previous subsection explained basic building blocks that can be used to understand any ideas of interaction in AI and organizations. Intelligence arises from learning to homeostatically “comport” one’s body and experience with situations as they unfold on a moment-to-moment basis. Actions for achieving such comportment are necessarily situated, embodied, and embedded. Symbol manipulation can be useful, but only secondarily as “resources for action” that help us “index” and “register” the external world as we experience it. What does all of this mean for actually designing an alternative sort of AI system or organization? Next, we will consider two main technical proposals: Brooks’ **subsumption architecture** (which does not require any symbol manipulation), and the notion of **direct manipulation** of symbols as resources for action.

4.1 Learning based on goals for interacting with the world: the subsumption architecture

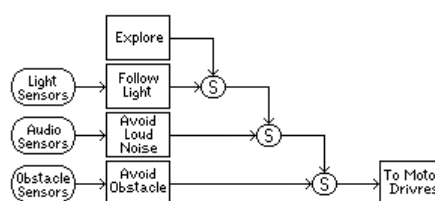
Brooks proposed that the sensorimotor actions fundamental to the evolution of intelligence could be understood as corresponding to **multiple goals** (also called **behaviors**) that need to be maintained by an agent. In this view, the challenge of intelligence is not so much to search a tree of actions or propagate information across a constraint network. Rather, the fundamental challenge of intelligence is to control (i.e., prioritize) goals (i.e., interactions) as situations change in the physical world. In organizations, a canonical recent example of this view of intelligence is the idea of a **digital ecosystem**, which has replaced the previous notion of a well-defined **industry** as the basic units in which strategic innovations take place. A digital ecosystem can be thought of as consisting of the various lower- and higher-level goals of a “**software stack**”. Lower-level goals are concern, for example, getting high-performing semiconductor chips; higher-level goals could be, for instance, engaging user interfaces. Intelligence is achieved by “**orchestrating**” all of the goals of the stack so that they work together.

Subsumption architecture. Brooks proposed that an AI system or organization could be formally designed as a set of **multiple goals** that correspond to **fundamental behaviors** to maintain in the physical world and informed by sense perception (using very simple sensor networks) rather than representations. The multiple goals in his proposal are learned and implemented in order (i.e., simulating “creature” evolution) from the most general ones to the more specific ones. The goals would then be given a **prioritization**, or what he labeled the ability to “**subsume**” other goals to control intelligent action on a moment-to-moment basis. For example, a robot would have the goal “do not run into walls” subsume other goals such as “explore the room” since the former goal has higher priority than the latter.

In contrast to constraint networks, the **goals or behaviors** of the subsumption architecture are **structured as layers**. Hence, the goals need not have explicit constraints that define how they are related. The key idea behind this AI system design, or what Brooks called the “**subsumption architecture**”, was that (1) having multiple layers of goals that are fundamental to action, (2) a strategy to learn them goal-by-goal, starting with the most fundamental, and (3) a way to control their priority for real-world action, makes an AI system or organization highly robust to changes in situations since the goals are so general that they will be relevant across these situations. Note there is no representation in Brooks’ idea in the sense that there is no real model of the world — what is represented is only the behavior that is desired from the agent’s *interaction* with the world. Further, these goals or behaviors in his proposal are only implemented through highly simple circuits, called finite state machines (FSMs), with no communication channels (i.e., this is not like Shannon information).



Basic elements of the subsumption architecture



Example architecture for a simple robot

4.2. Direct manipulation of symbols

Another main area of research on interaction in AI and organizations concerns directly interacting with physical media, and specifically digital representations in personal computers and other personal or collaborative digital devices. The key insight in this area of research is that, *even* when we explicitly use symbolic representations, we interact with them through sensorimotor actions and on physical devices in particular situations. For example, we take physical actions (e.g., swiping, clicking, viewing) to manipulate screens on our laptops, tablets or phones. Businesses today frequently depend on designing these physical interactions with digital

representations — what we will call **direct manipulation** of information — to be intuitive and “seamless”. Apple’s famous graphical user interfaces, for example, were fundamental to its value proposition of enabling consumers to interact flexibly both within and across many software applications. Ridesharing services such as Uber or Kakao Taxi depend on interactively providing relevant information about location and prices in real-time on a smartphone screen. In short, we access digital representations through **interactive media**.

The background for this perspective comes in large part from work by **engineers**, and specifically **electrical engineers** from **MIT** and a few other institutions. In the 1950s and 1960s at MIT, much pioneering work in electrical engineering was being done, such as designing and building digital circuits and other hardware for digital computers and networks. Most of this work did not have any obvious division of labor for specializing tasks, as in the earlier pin factory example that we considered. A team of engineers must reason about an *entire* circuit, or at least a large section of it. As in any engineering, this reasoning is largely based on “tinkering”, or physically manipulating the components of whatever is being built, to “debug” flaws in the design that could not be discovered without physical testing.

The electrical engineers at MIT also had access, however, to some of the earliest digital computers, and some of these engineers became pioneers in the AI field. Their focus was on how symbols, though abstract in one sense, in fact need to be interacted with in the physical world to solve complex problems. Just like an engineer (or architect) likes to tinker through constructing and directly manipulating an actual physical system (e.g., by changing the physical components, or by drafting the design of a machine with pen and paper), the idea was that an engineer could directly manipulate the components as graphical objects on a physical screen — they could tinker in the physical space of a computer screen.

The electrical engineering work involved solving complex problems by constructing and reasoning about schematic circuit diagrams to capture the physical properties of the circuits (or other devices). The budding AI scholars, epitomized by **Ivan Sutherland’s Sketchpad** system (the forerunner to CAD software and the GUI (graphical user interface) of personal computers), envisioned software that would enable drafting a diagram of a circuit or other complex machine in digital form, and then “tinkering” with this by interacting with the diagram on a physical screen. Here, the problems could be understanding the “flow of current” through a circuit, and the performance goals could be based on physical laws of electrical engineering such as the maximum current for a given circuit. The idea was that **direct manipulation** of graphical objects on a digital computer would be more flexible than tinkering with the actual physical circuit, but also much simpler and more flexible than writing sequences of code to describe a circuit. Analogous to McCarthy’s ideas about programming as declarative, the engineer only needed to directly manipulate a computer since the computer stored constraints about what the circuit should do, rather than precisely *how* to do it.